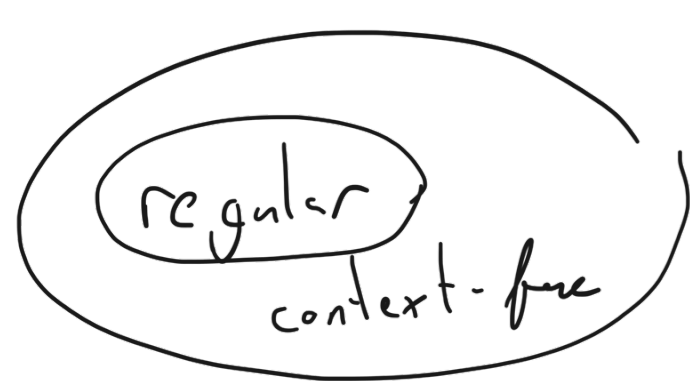


Context-free grammars and languages



context-free grammar

every language described by a CFG

Grammar G_1 :

$A \rightarrow OA$
 $A \rightarrow B$
 $B \rightarrow \#$

variables: A, B
terminal: #
collection of substitution rules (productions): $A \rightarrow OA, A \rightarrow B, B \rightarrow \#$

To generate a string.

1. Write down the start variable, which is in the top-left most rule.
2. Replace a variable with a production.
3. Repeat step 2 until only terminals remain.

Generates: 000#111

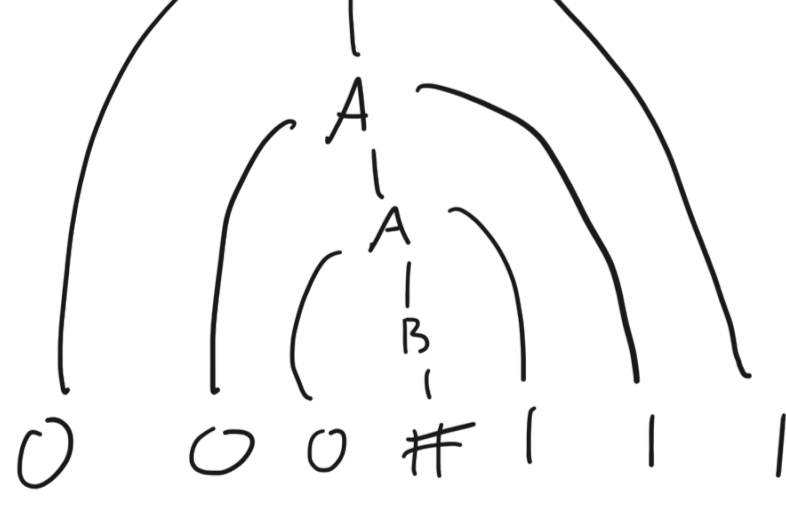
$A \Rightarrow OA \Rightarrow OOA \Rightarrow OOOA \Rightarrow OOOB \Rightarrow OOO\#111$

$A^* \Rightarrow OOO\#111$

yields/derives

derivation

Can show a parse tree



Formal Def A context-free grammar (CFG) is a 4-tuple (V, Σ, R, S) where

1. V is the finite set of variables
2. Σ is the finite set of terminals s.t. $\Sigma \cap V = \emptyset$.
3. R is the set of rules, each rule being a variable and a string of variables and terminals
4. $S \in V$ is the start state.

Ex 1 $L_1 = (0 \cup 1)^*$ CFG:

$S \rightarrow SO \mid S1 \mid \epsilon$

$L(S) = L_1$

$w = 001$

$S \Rightarrow \underline{S}1 \Rightarrow \underline{S}O1 \Rightarrow \underline{S}OO1 \Rightarrow \epsilon OO1 \Rightarrow 001$

Ex 2 $A = \{0^n 1^n \mid n \geq 0\} \cup \{1^n 0^n \mid n \geq 0\}$

$S \rightarrow S_1 \mid S_2$

$S_1 \rightarrow OS_11 \mid \epsilon$

$S_2 \rightarrow 1S_20 \mid \epsilon$

$B = \{0^n 1^m \mid n, m \geq 0\}$

$= 0^* 1^*$

$S \rightarrow OS \mid P$

$P \rightarrow 1P \mid \epsilon$

001

$S \Rightarrow OS \Rightarrow OOS \Rightarrow OOP \Rightarrow$

$OO1P \Rightarrow 001$

$S \Rightarrow 001$

Operations on CFGs

Let S_1 & S_2 be the starting variables of CFGs L_1 & L_2 .

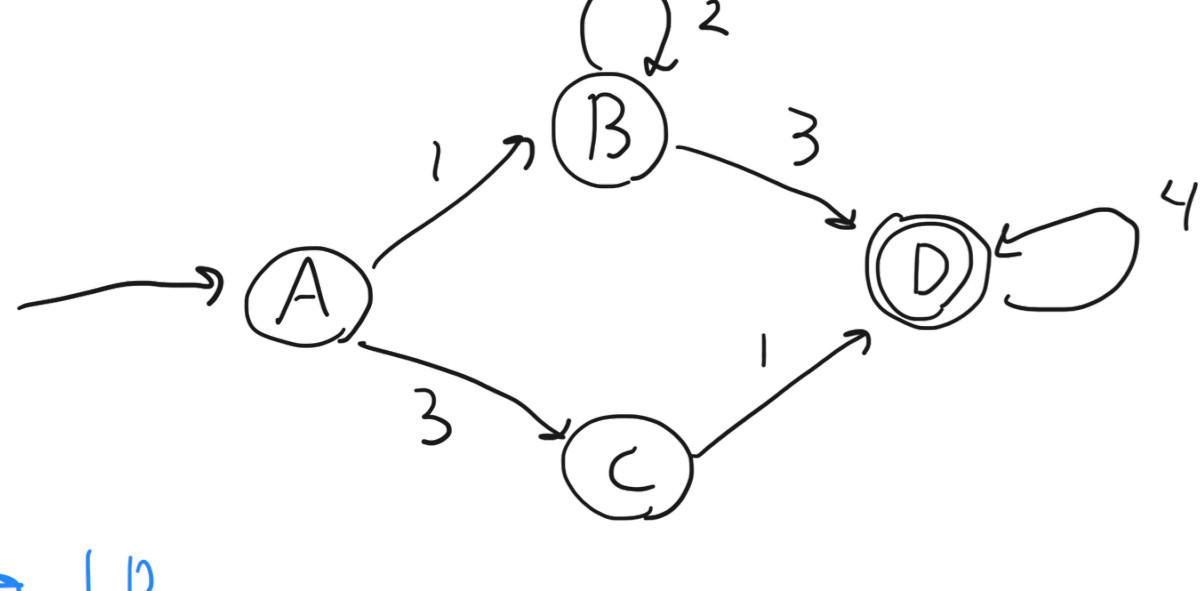
$L_1 \cup L_2 \Rightarrow S \rightarrow S_1 \mid S_2$

$L_1 \circ L_2 \Rightarrow S \rightarrow S_1 S_2$

$L_1^* \Rightarrow S \rightarrow \epsilon \mid S_1 S$

Theorem: Every regular language is context-free

- Make a variable for each state
- Make a rule for each edge.
- Add an epsilon rule for each accept state.



$A \rightarrow 1B$

$A \rightarrow 3C$

$B \rightarrow 2B$

$B \rightarrow 3D$

$C \rightarrow 1D$

$D \rightarrow 4D$

$D \rightarrow \epsilon$

$P = \{ \text{palindromes over } \{0,1\} \}$

00
010
01010

$S \rightarrow \epsilon \mid 0 \mid 1 \mid OSO \mid 1S1$

$S \Rightarrow OSO \Rightarrow O1S1O \Rightarrow 01010$

$W = \{ w \mid w \in \{0,1\}^* \text{ and the number of 0's equals the number of 1's} \}$

$S := OA \mid 1B \mid \epsilon$ yields strings with equal 1's & 0's.

$A := 1S \mid OAA$

$B := OS \mid 1BB$

yields strings with one more 1 than 0's.

yields strings with one more 0 than 1's.

00010111

$S \Rightarrow OA \Rightarrow OAAA \Rightarrow OOOAAA \Rightarrow OOO1SAA$

$\Rightarrow OOO1OAAA \Rightarrow \dots \Rightarrow OOO1O1S1S1S \Rightarrow \dots \Rightarrow OOO1O111$

$S := SAB \mid \epsilon$

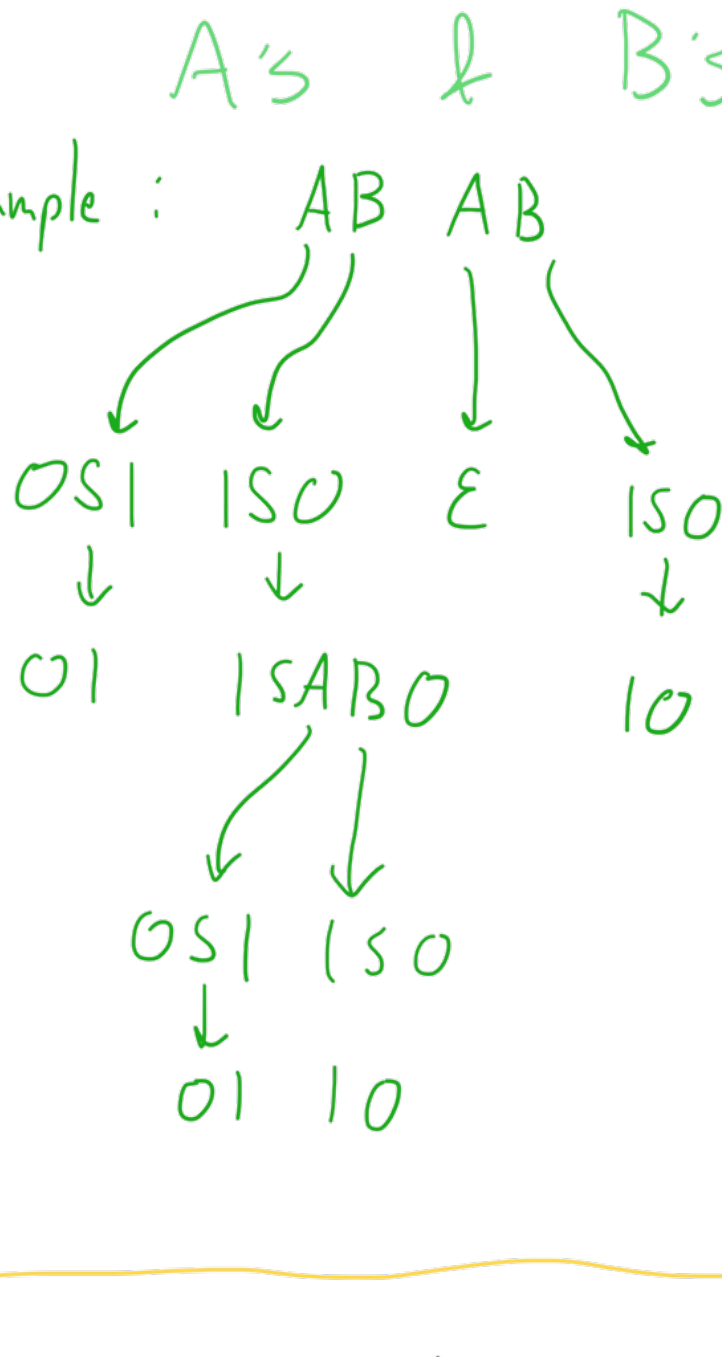
$A := OS1 \mid \epsilon$

$B := 1SO \mid \epsilon$

$AB^* = AB AB AB \dots$

Every AB can go to ϵ or any string of A's & B's.

For example:



No substitution that results in more 0's than 1's or vice versa is possible!

A grammar is ambiguous if it generates some string via 2 or more derivations.

$\langle \text{EXPR} \rangle \rightarrow \langle \text{EXPR} \rangle + \langle \text{EXPR} \rangle \mid \langle \text{EXPR} \rangle \times \langle \text{EXPR} \rangle \mid \langle \text{EXPR} \rangle \mid a$

